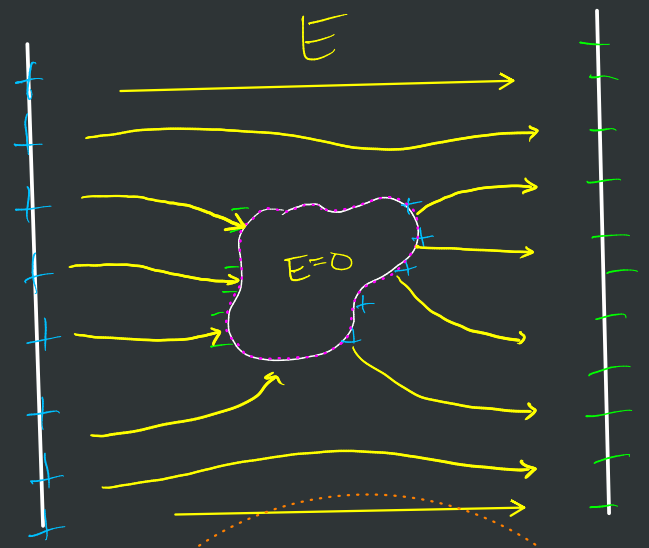
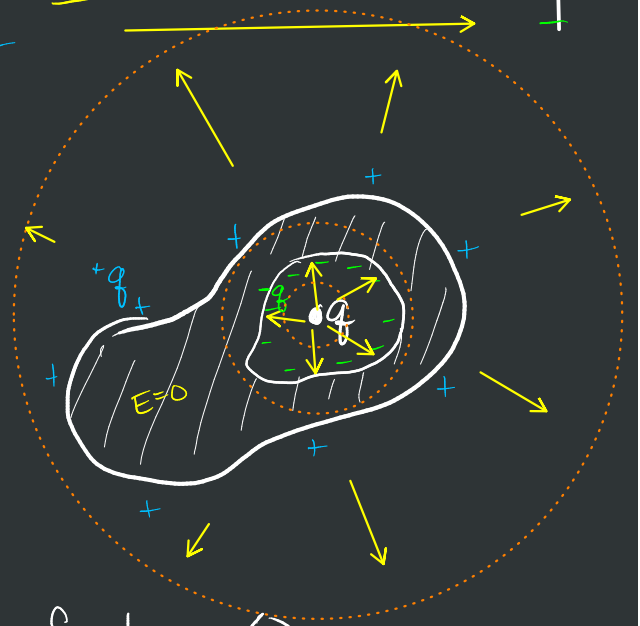
Chapter 3

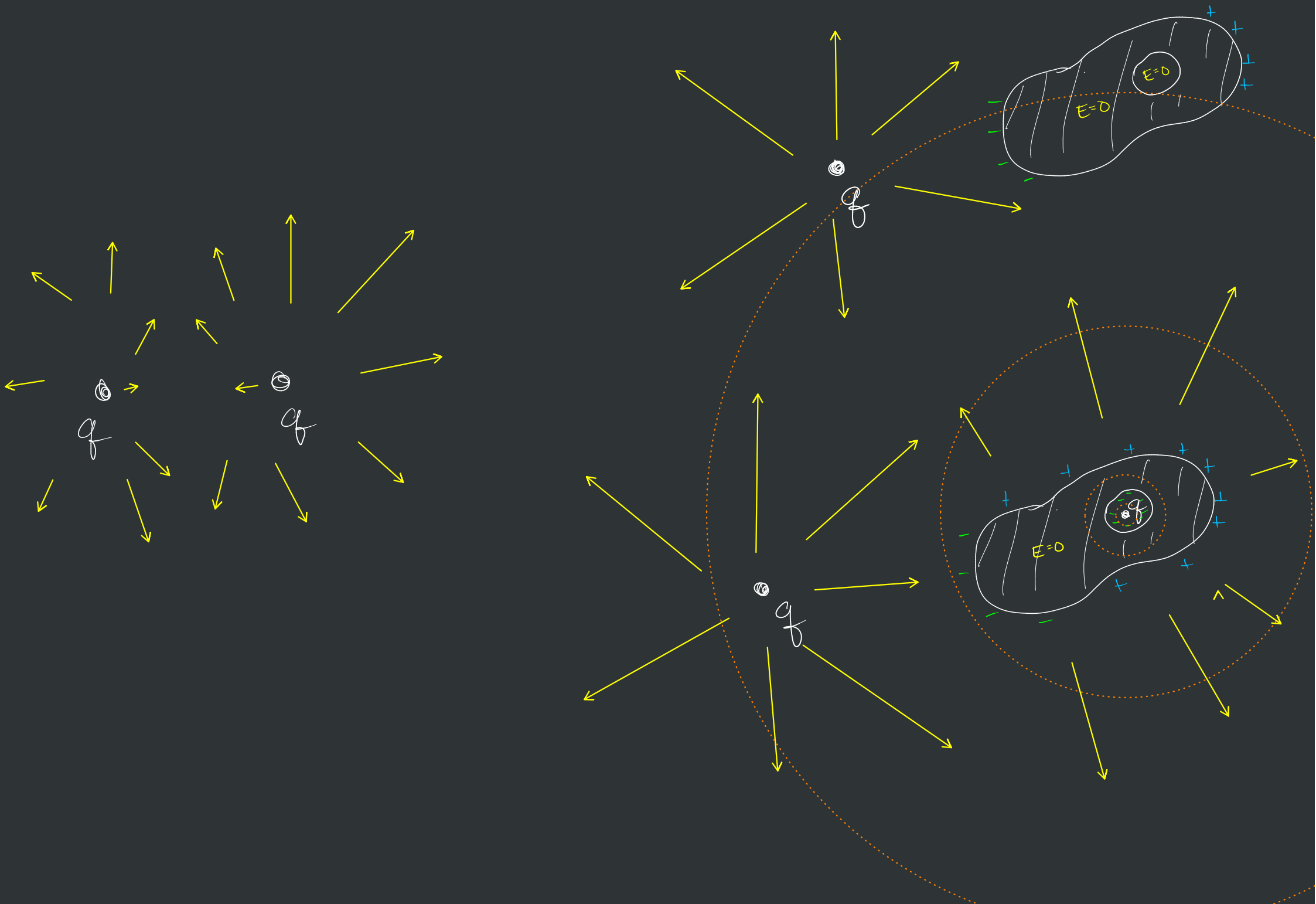
Conductors vs. **insulators** → shift of charges but mass movement of charge.

→ charges move freely (no work) through the bulk of the material



- in the presence of an electric field charges move w/i the conductor until the field inside is zero.
- so the charge reside on the surface
- so the surface is an equipotential
- electric field just outside the conductor is perpendicular to the surface
- at the surface Gauss's Law is satisfied for the total electric field $\epsilon_0 \oint_S \vec{E} \cdot d\vec{a} = \int \sigma \, da = Q_{NET}$




q
q
q
E=0
E=0
q
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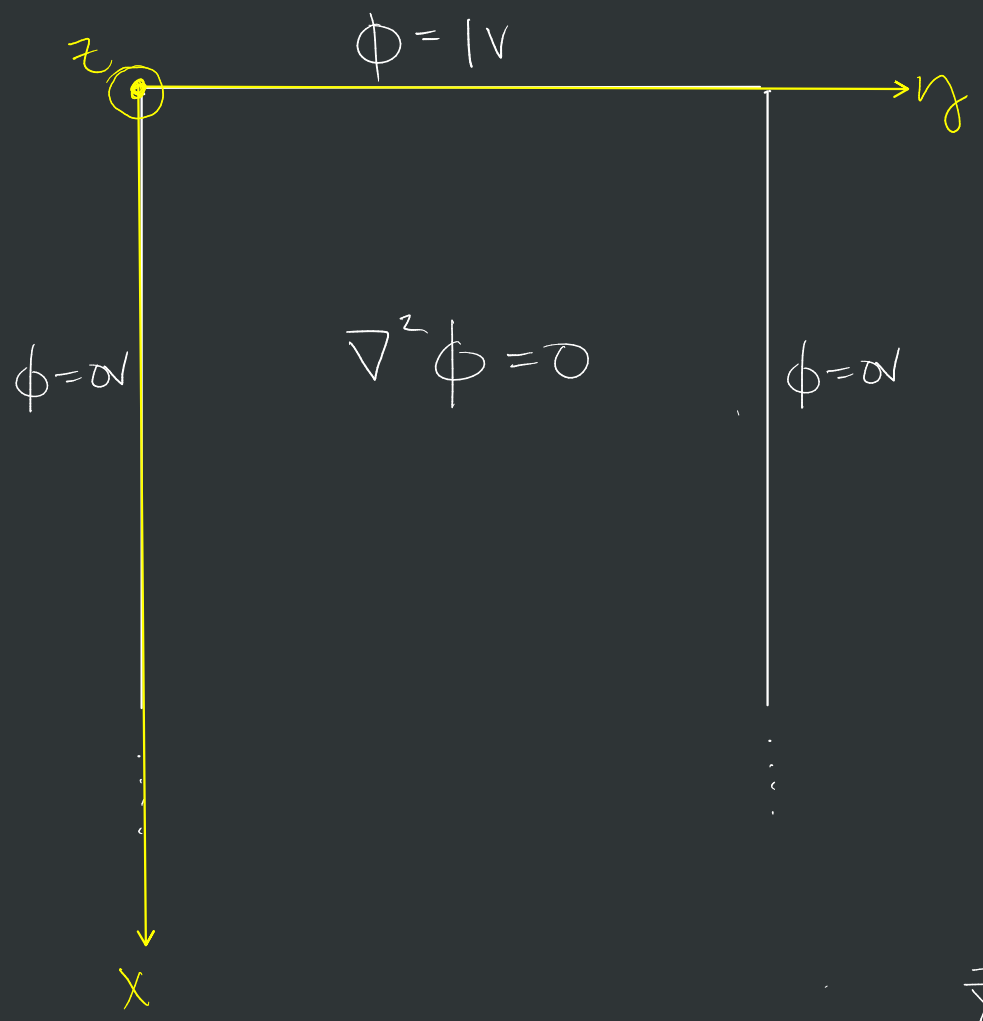


in cartesian

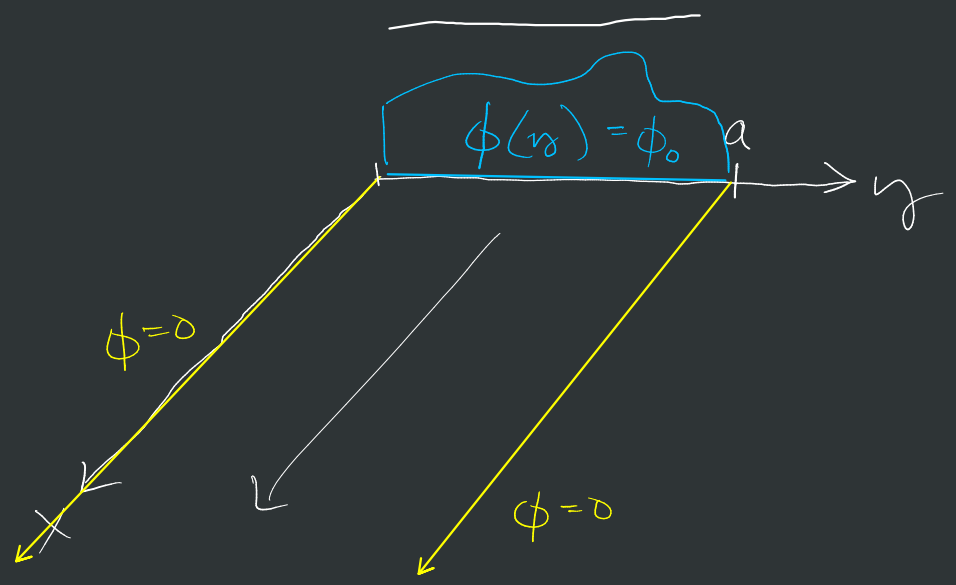
$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \underbrace{\frac{\partial^2 \phi}{\partial z^2}}_{=0} = 0$$

$$\phi = X(x) \cdot Y(y) \leftarrow$$

$$\frac{\partial^2 X(x) Y(y)}{\partial x^2} + \frac{\partial^2 X Y}{\partial y^2} = 0$$

$$\frac{1}{X \cdot Y} \left[Y \frac{\partial^2 X}{\partial x^2} + X \frac{\partial^2 Y}{\partial y^2} = 0 \right] \frac{1}{X \cdot Y}$$

$$\underbrace{\frac{1}{X} \frac{\partial^2 X}{\partial x^2}}_{= -k^2} + \underbrace{\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2}}_{= k^2} = 0$$



$$\frac{1}{X}\frac{d^2X}{dx^2} = +k^2 \qquad \frac{1}{Y}\cdot\frac{d^2Y}{dx^2} = -k^2$$

$$\underbrace{\frac{d^2X}{dx^2} = +k^2 X}_{X(x) = Ae^{kx} \text{ or } Be^{-kx}} \qquad \underbrace{\frac{d^2Y}{dy^2} = -k^2 Y}_{Y(y) = C\sin(ky) \text{ or } = D\cos(ky)}$$

so $X(x) = Ae^{kx} + Be^{-kx}$

$Y(y) = C\sin(ky) + D\cos(ky)$

$$\phi = XY$$

$x = 0, \ \phi = \phi_0$ | ✓ $y = 0, \ \phi = 0$

✓ $x = \infty, \ \phi = 0$ | ✓ $y = a, \ \phi = 0$

$X(x) = A\cdot e^{kx} + Be^{-kx}$. $Y(y) = C\sin(ky) + \underbrace{D\cos(ky)}_{D=0}$

$A = 0$

$X(x) = Be^{-kx}$

$Y(y) = C\sin(ky)$

$$\phi(x,y) = Be^{-kx}\cdot C\sin(ky)$$

$$\phi(x,y) = Be^{-kx}\sin(ky)$$

$$\phi(x,a) = 0 = Be^{-kx}\sin(ka)$$

$$0 = \sin(ka)$$

$$ka = \underbrace{\pi, 2\pi, 3\pi, 4\pi, \ldots}_{n\pi}$$

$$k = \frac{n\pi}{a} \quad \text{for integer } n$$

$$\boxed{\phi(x,y) = \sum_{n=1}^{\infty} C_n e^{-\frac{n\pi}{a}x}\sin\left(\frac{n\pi}{a}y\right)}$$

use the last B.C.

$$x=0 \quad \phi = \phi(y)$$

$$\phi(0,y) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{a}y\right)$$

multiply both sides by $\sin\left(\frac{n'\pi}{a}y\right)$

$$\nabla^2\phi = 0$$

$$\alpha_1\nabla^2\phi_1 + \alpha_2\nabla^2\phi_2 + \ldots = 0$$

$$\nabla^2(\alpha_1\phi_1 + \alpha_2\phi_2 + \ldots) = 0$$

$$\phi \sin\left(\frac{n'\pi}{a} y\right) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{a} y\right) \sin\left(\frac{n'\pi}{a} y\right)$$

$$\int_0^a \phi(0,y) \sin\left(\frac{n'\pi}{a} y\right) dy = \sum_{n=1}^{\infty} C_n \underbrace{\int_0^a \sin\left(\frac{n\pi}{a} y\right) \sin\left(\frac{n'\pi}{a} y\right) dy}$$

$n = n'$

$\to = \frac{a}{2}$

$n \neq n'$

$\to = 0$

$$\int_0^a \phi(0,y) \sin\left(\frac{n'\pi}{a} y\right) dy = C_{n'} \cdot \frac{a}{2}$$

$$C_{n'} = \frac{2}{a} \int_0^a \phi(0,y) \sin\left(\frac{n'\pi}{a} y\right) dy$$

drop primes

$$\boxed{C_n = \frac{2}{a} \int_0^a \phi(0,y) \sin\left(\frac{n\pi}{a} y\right) dy}$$

Suppose $\phi(0,y) = \phi_0$ ← constant potential boundary

$$C_n = \frac{2}{a}\int_0^a \phi_0 \sin\left(\frac{n\pi}{a}y\right)dy$$

$$C_n = \frac{2\phi_0}{a}\underbrace{\int_0^a \sin\left(\frac{n\pi}{a}y\right)dy}$$

$u = \frac{n\pi}{a}y$ $\quad du = \frac{n\pi}{a}dy$

$\frac{a\,du}{n\pi} = dy$

$$C_n = \frac{2\phi_0}{\cancel{a}}\left[-\frac{\cancel{a}}{n\pi}\cos\left(\frac{n\pi}{a}y\right)\Big|_0^a\right]$$

$\int \sin u \left(\frac{a}{n\pi}\right)du$

$-\cos(u)$

$$= -\frac{2\phi_0}{n\pi}\left[\underbrace{\cos(n\pi)}_{} - \underbrace{\cos(0)}_{1}\right]$$

1 $\quad n$ is even

-1 $\quad n$ is odd

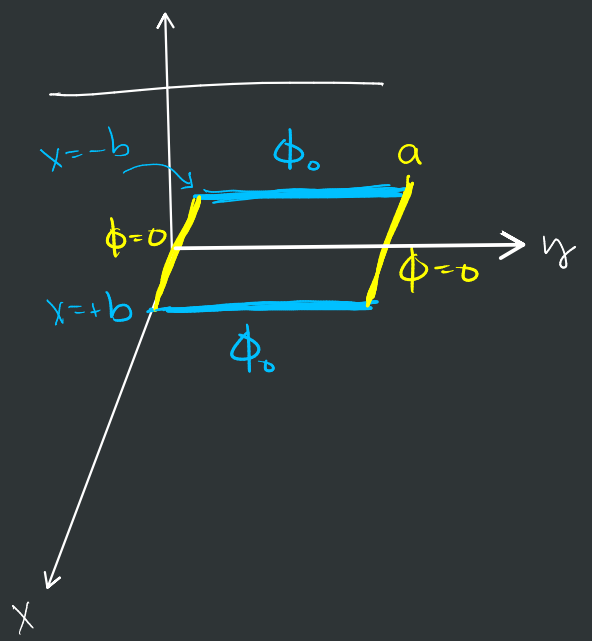
$$C_n = \frac{4\phi_0}{n\pi} \quad \text{when } n \text{ is odd}$$

$$= 0 \quad \text{when } n \text{ is even}$$

$$\phi(x,y) = \sum_{n=1}^{\infty} C_n e^{-\frac{n\pi}{a}x}\sin\left(\frac{n\pi}{a}y\right)$$

$$\boxed{\phi(x,y) = \frac{4\phi_0}{\pi} \sum_{n \text{ is odd}}^{\infty} \frac{1}{n} e^{-\frac{n\pi}{a}x} \cdot \sin\left(\frac{n\pi}{a} y\right)}$$

$$= \frac{4\phi_0}{\pi} \sum_{n=0}^{\infty} e^{-\frac{(2n+1)\pi}{a}x} \cdot \sin\left(\frac{(2n+1)\pi}{a} y\right)$$



$$\underbrace{\frac{1}{X}\frac{\partial^2 X}{\partial x^2}}_{+k^2} + \underbrace{\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2}}_{-k^2} = 0$$

$$\frac{\partial^2 X}{\partial x^2} = +k^2 X \qquad \frac{\partial^2 Y}{\partial y^2} = -k^2 Y$$

$k^2 > 0 \rightarrow$ exponentials

$$X = Ae^{kx} + Be^{-kx}$$

$$\phi_0 = Ae^{kb} + Be^{-kb}$$

$$\left.\begin{aligned} \phi_0 &= Ae^{-kb} + Be^{kb} \\ \phi_0 &= Be^{kb} + Ae^{-kb} \end{aligned}\right\} A = B$$

$k^2 < 0 \rightarrow$ sin/cos

$$Y = C\sin(kx) + D\cos(kx)$$

$y=0, \phi=0$ $\qquad$ $y=a, \phi=0$

$$0 = \underbrace{C\sin(0)}_{0} + \underbrace{D\cos(0)}_{D=0}$$

$$\phi(x,y) = X(x)\cdot Y(y) = A\left(e^{kx} + e^{-kx}\right)\cdot C\sin(ky) \quad \leftarrow \text{so far}$$

$$\phi(x,y) = C\left(e^{kx} + e^{-kx}\right)\sin(ky)$$

$$0 = C\left(e^{kx} + e^{-kx}\right)\sin(ka)$$

$$0 = \sin(ka)$$

$$ka = n\pi$$

$$k = \frac{n\pi}{a} \qquad n = 1, 2, 3, \ldots$$

$$\phi(x,y) = \sum_{n=1}^{\infty} C_n\left(e^{\frac{n\pi}{a}x} + e^{-\frac{n\pi}{a}x}\right)\sin\left(\frac{n\pi}{a}y\right)$$

find C_n